Geometry & Coordinate Geometry

Section 1: Vectors & Dot Products

1 What is the dot product of the vectors $u = \langle 3, -4 \rangle$ and $v = \langle 2, 5 \rangle$?

a. -14

b. 26

c. 14

d. -26

2 For what value of k are the vectors $u = \langle k, 3 \rangle$ and $v = \langle 6, -2 \rangle$ perpendicular?

a. 1

b. 2

c. -1

d. 3

3 What is the magnitude (length) of the vector $v = \langle 5, -12 \rangle$?

a. 13

b. 17

c. 7

d. 169

Section 2: Distance, Midpoint & Lines

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4 What is the distance between the points $A(1, 2)$ and $B(4, 6)$?

a. 5

b. 7

c. $\sqrt{7}$

d. 25

5 What are the coordinates of the midpoint of the segment joining $P(-2, 7)$ and $Q(6, 1)$?

a. $(2, 4)$

b. $(4, 8)$

c. $(2, 3)$

d. $(1, 4)$

6 What is the slope of a line perpendicular to the line $3x - 4y = 8$?

a. $-\frac{4}{3}$

b. $\frac{3}{4}$

c. $\frac{4}{3}$

d. $-\frac{3}{4}$

Section 3: Conic Sections (Parabola & Circle)

7 What is the center and radius of the circle $(x - 3)^2 + (y + 2)^2 = 36$?

a. Center $(3, -2)$, radius 6

b. Center $(-3, 2)$, radius 6

c. Center $(3, -2)$, radius 36

d. Center $(-3, 2)$, radius 36

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8 In the parabola $(y-1)^2 = 8(x+2)$, which direction does it open?

a. To the right

b. To the left

c. Upward

d. Downward

9 What is the focal length p of the parabola $x^2 = 12y$?

a. 3

b. 6

c. 12

d. 4

Section 4: Conic Sections (Ellipse & Hyperbola)

10 What is the length of the major axis of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$?

a. 10

b. 5

c. 8

d. 20

11 What is the eccentricity e of a conic section if it represents a hyperbola?

a. $e > 1$

b. $e = 1$

c. $0 < e < 1$

d. $e = 0$

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12 What type of conic section is represented by $x^2 - 4y^2 - 2x + 16y - 19 = 0$?

a. Hyperbola

b. Ellipse

c. Parabola

d. Circle

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